Human Classifier Simulation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Human Classifier Simulation before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Features and Distractors

Without opening the old lesson, explain one key idea from Features and Distractors. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Rows Columns and Data Types

Without opening the old lesson, explain one key idea from Rows Columns and Data Types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Trace Tables

Without opening the old lesson, explain one key idea from Trace Tables. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Human Classifier Simulation trace and record the first step you can explain without help.

Use W2: Construct both outcome paths for Constructing a **classifier**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Human Classifier Simulation explanation against the supplied facts before discussing it.

Use the worked case for Constructing a **classifier**. Proposed explanation: “Any numeric **input** must receive a category”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

A **written rule** selects science books with fewer than 140 pages. Trace science/139, science/140, history/90 and missing-genre/90. Policy: missing required information means request it.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Human Classifier Simulation target to a trace and a justified conclusion.

Recommend a book only if genre is science AND pages<=150. Classify science/150, science/151, mystery/90 and missing-genre/80. State a missing-data policy.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Human Classifier Simulation case and show the deciding evidence.

Recommend if adventure AND pages<150; otherwise Other.

Trace A=(adventure,120), B=(adventure,180), C=(science,90).

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

A human **classifier** uses sports topic means sports shelf. Classify a sports book and an unreadable cover. State what **evidence** is missing in the second case.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 10 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, **boundary** case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Human Classifier Simulation adds to Transparent Book Recommender and keep the evidence.

Turn score thresholds into visible boundaries such as recommend / maybe / not now.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Recommend a book only if genre is science AND pages<=150. Classify science/150, science/151, mystery/90 and missing-genre/80. State a missing-data policy.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Write an input contract

> Write an input contract. Apply the supplied Human Classifier Simulation facts and keep your reasoning.

Write the complete Tower1 **input contract**. Give one accepted example and three different rejected examples, each with its reason. Explain why an accepted **input** does not have to be one of the examples used to fit the threshold.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Construct both outcome paths

> Construct both outcome paths. Apply the supplied Human Classifier Simulation facts and keep your reasoning.

Write numbered instructions for Tower1, including **input** checks, the stopping path after a failed check, the threshold rule and the record to keep. Your partner must be able to use them without supplying an unstated rule.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Trace boundaries and a broken path

> Trace boundaries and a broken path. Apply the supplied Human Classifier Simulation facts and keep your reasoning.

Trace 1 cm, 5 cm, 6 cm, 10 cm and 11 cm using Tower1. A faulty procedure records **no prediction** for a blank, then continues and treats the blank as zero to report short.

Identify the missing instruction and the proper final outcome.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Build an independent ribbon classifier

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Build an independent ribbon classifier. Apply the supplied Human Classifier Simulation facts and keep your reasoning.

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise.

Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for **no prediction** and the **model version** in each record.

My reasoning and evidence

W5 Explain what your procedure establishes

> Explain what your procedure establishes. Apply the supplied Human Classifier Simulation facts and keep your reasoning.

Ribbon1 accepts 12 cm and predicts long. No comparison label has been supplied for that record. State what has been checked and what remains unknown.

Explain why changing a blank to 1 cm just to produce a category would be a mistake.

My reasoning and evidence

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Human Classifier Simulation

> A fictional librarian wants illustrated mystery books with at most 100 pages. Explain the labelled step before checking the changed condition.

A fictional librarian wants illustrated mystery books with at most 100 pages.

Written rule: choose only genre mystery AND illustrated True AND pages<=100.

Rows J=(mystery,True,100), K=(mystery,False,90), L=(science,True,80), M=(mystery,True,blank). Missing required fields request review.

AI ACADEMY / AI-2/10

Human Classifier Simulation

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Read all conditions

Three requirements must pass, not just a high preference score.

2. Trace J

Mystery, illustrated and exactly 100 pages: choose.

3. Trace K and L

K fails illustration; L fails genre: do not choose.

4. Trace M

Missing pages means review rather than assuming it is short.

CHANGE ONE THING

Supply M's pages as 101.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why are J and M initially treated differently even though both are illustrated mysteries?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional librarian wants illustrated mystery books with at most 100 pages.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Missing pages means review rather than assuming it is short.

1. Read all conditions

2. Trace M

3. Trace J

4. Trace K and L

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Trace M Missing pages means review rather than assuming it is short. This human simulation lets students inspect every decision without needing a trained model.

Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check. The order should not turn blank into zero or quietly ignore a required field.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Supply M's pages as 101. Which conclusion is supported by the supplied facts?

1. M is now rejected rather than awaiting page review.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: M is now rejected rather than

awaiting page review. The required field is known but fails the maximum-length condition.

Repair a missing building block

Review Features and Distractors (ai-2/9). Goal: Choose recorded properties that preserve distinctions needed for a stated task. Short practice: Does this one pair prove the **classifier** always ignores subject?

Explain a narrower claim.. Return to this week and explain what you changed.

Review Rows Columns and Data Types (ai-2/7). Goal: Specify a table schema. Short practice: Which two rows originally fail, and do they fail for the same reason?. Return to this week and explain what you changed.

Review Trace Tables (ai-2/4). Goal: Trace variable values before and after each loop iteration. Short practice: Move print inside the original loop. What sequence is displayed?.

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

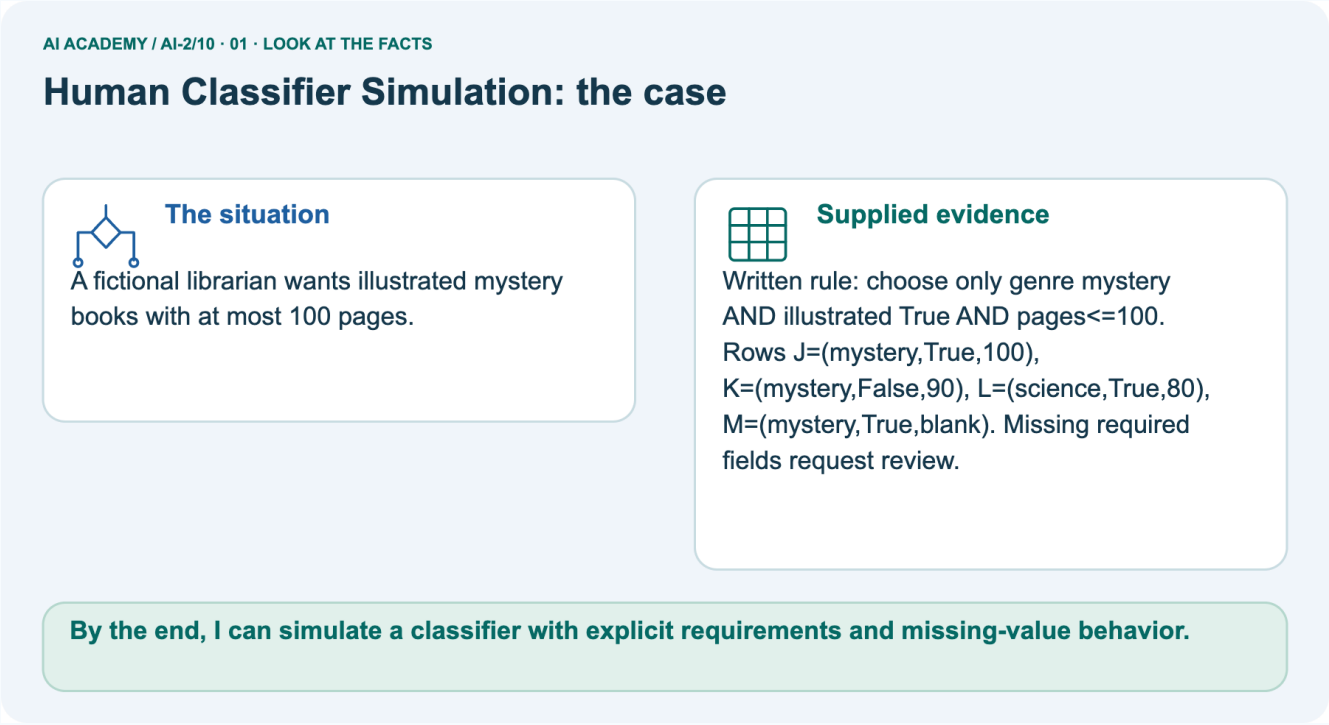
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Human Classifier Simulation picture from its labels, then record a relevant limit.

01. Human Classifier Simulation: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

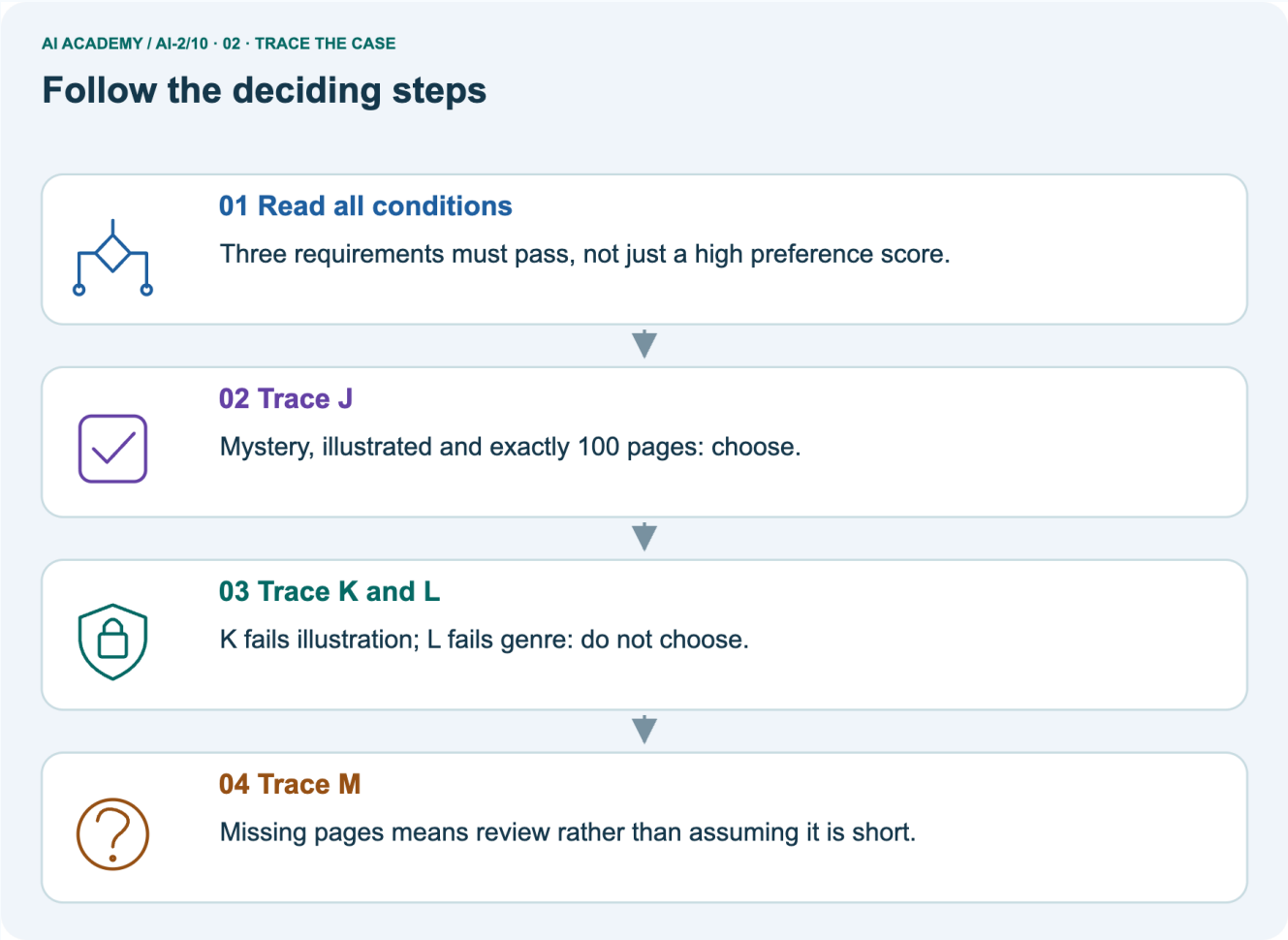
A fictional librarian wants illustrated mystery books with at most 100 pages. Supplied facts: **Written rule**: choose only genre mystery AND illustrated True AND pages<=100.

Rows J=(mystery,True,100), K=(mystery,False,90), L=(science,True,80), M=(mystery,True,blank). Missing required fields request review.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



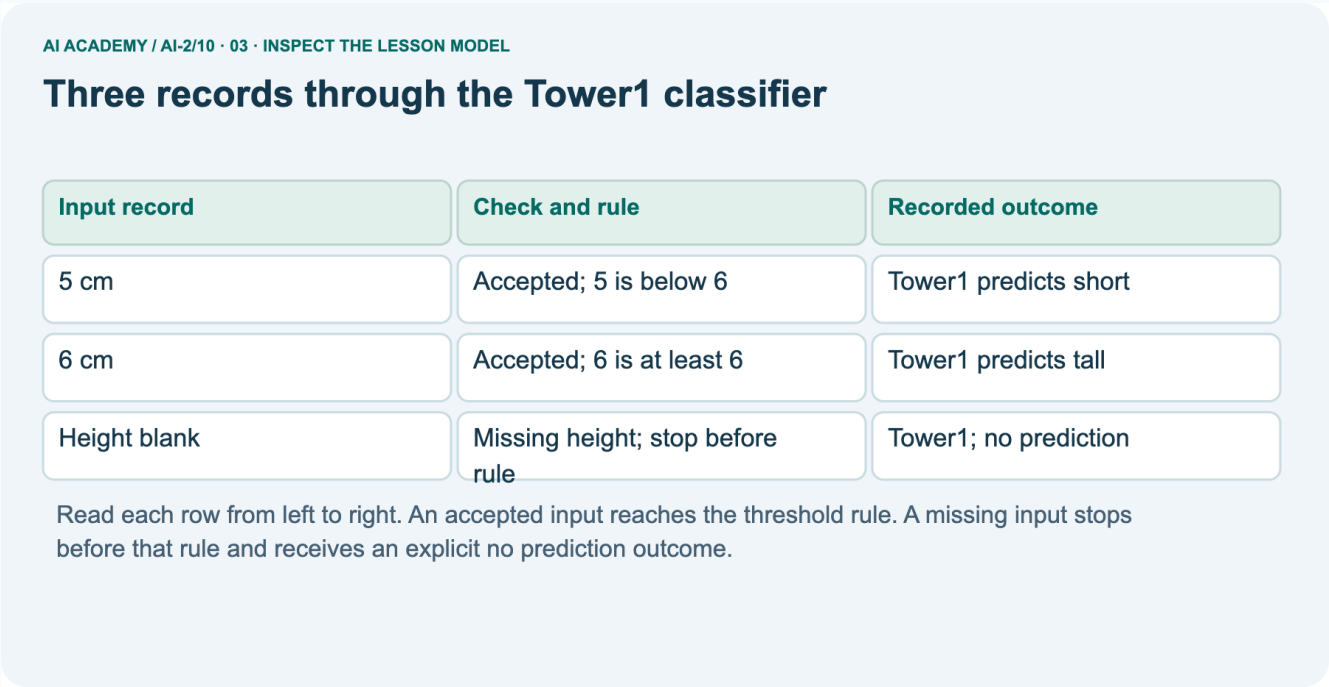
Follow the labelled picture with the matching explanation. Open the vector diagram

Read all conditions: Three requirements must pass, not just a high preference score. → Trace J: Mystery, illustrated and exactly 100 pages: choose. → Trace K and L: K fails illustration; L fails genre: do not choose. → Trace M: Missing pages means review rather than assuming it is short.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Three records through the Tower1 classifier



Follow the labelled picture with the matching explanation. Open the vector diagram

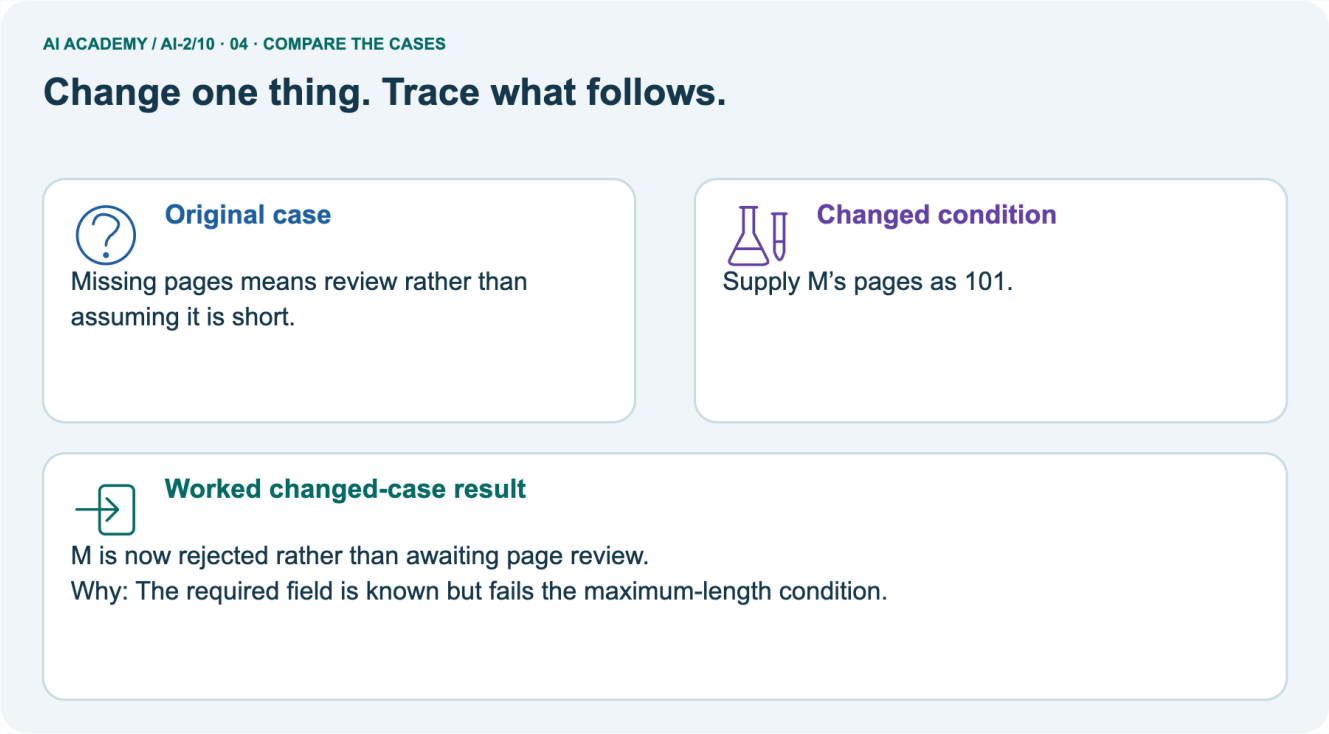
Three records through the Tower1 **classifier**. 5 cm | Accepted; 5 is below 6 | Tower1 predicts short. 6 cm | Accepted; 6 is at least 6 | Tower1 predicts tall.

Height blank | Missing height; stop before rule | Tower1; **no prediction**

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

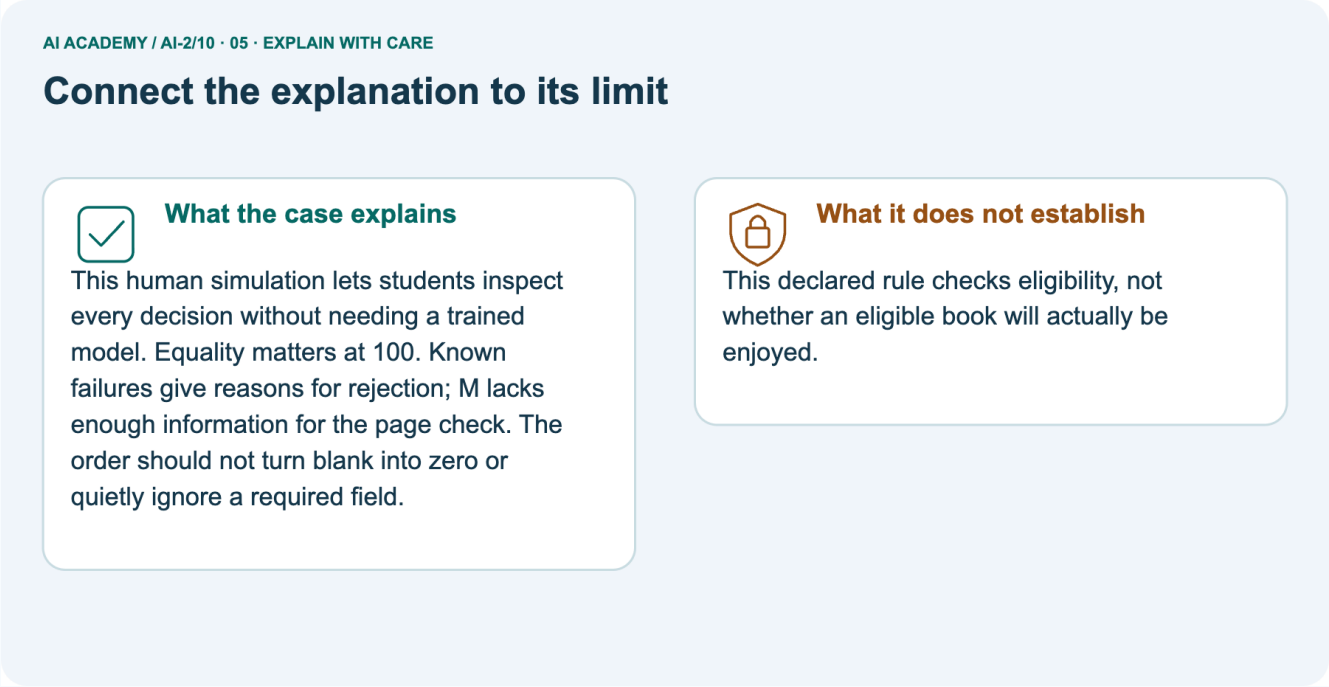
Original case: Missing pages means review rather than assuming it is short. Changed condition: Supply M's pages as 101. Expected result: M is now rejected rather than awaiting page review.

Reason: The required field is known but fails the maximum-length condition.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

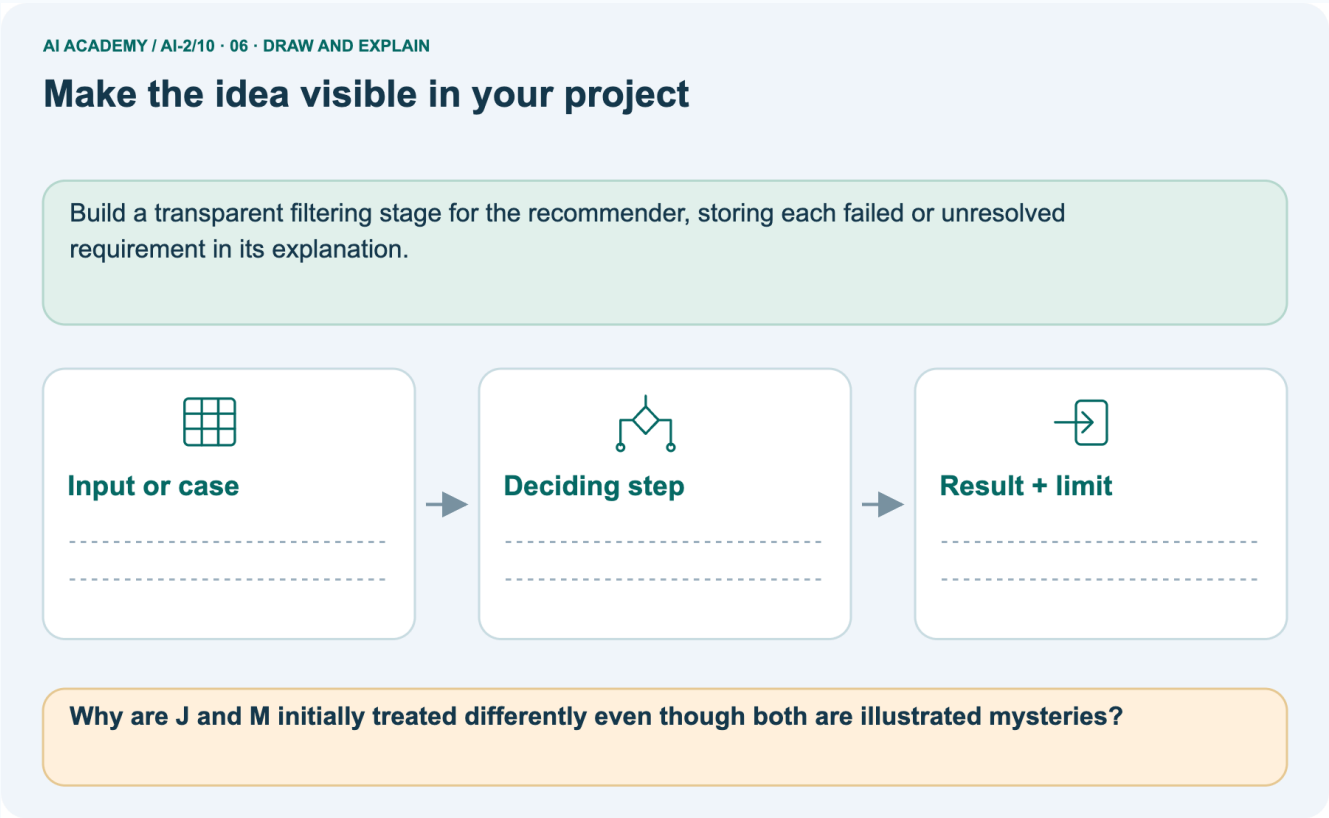
This human simulation lets students inspect every decision without needing a trained model. Equality matters at 100. Known failures give reasons for rejection; M lacks enough information for the page check.

The order should not turn blank into zero or quietly ignore a required field. Limit: This declared rule checks eligibility, not whether an eligible book will actually be enjoyed.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Build a transparent filtering stage for the recommender, storing each failed or unresolved requirement in its explanation. Draw your **input**, deciding step, result and limit.

Supported question: Why are J and M initially treated differently even though both are illustrated mysteries?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.